

Project Proposal: Mitigating Dialect Bias in Toxicity Detection

Research Question

How can we build a toxicity detector that accurately identifies harmful language without unfairly flagging African American English (AAE) as toxic?

This project will specifically try to investigate whether algorithmic "in-processing" modifications, such as custom loss functions and hyperparameters, can distinguish between sociocultural linguistic markers and actual toxic intent. Specifically, we aim to close the fairness gap in automated moderation systems without degrading overall toxicity detection performance.

Motivation

Automated moderation systems are widely used across social platforms to detect and remove toxic or abusive content. However, prior research has shown that these systems frequently exhibit dialectal bias, disproportionately flagging African American English (AAE) as toxic. Because most training datasets are dominated by Standard American English (SAE), models often learn spurious correlations between AAE linguistic markers, such as slang or reclaimed terms, and toxicity. This has been described as "linguistic racism," where algorithmic moderation disproportionately suppresses speech from certain communities. Empirical studies indicate that AAE-aligned text can receive toxicity scores 1.8 times higher and identity-hate scores up to 8.8 times higher than SAE text ([Ghimire 2025](#)). This leads to digital silencing, where minority voices are censored more often than others by mistake. Such disparities raise important questions about the fairness and reliability of machine learning systems deployed at scale.

Datasets

Blodgett Twitter AAE Corpus: Provides probabilistic labels indicating whether tweets align with African American English (AAE) or Mainstream American English (MAE). These scores will be used to construct proxy dialect labels.

Davidson Hate Speech Dataset: Contains labeled tweets categorized as hate speech, offensive language, or neither.

HateXplain: Provides hate speech annotations along with human rationales explaining why a piece of text is considered harmful.

Implementation Plan

Baseline 1: Standard XGBoost

We will train a Gradient Boosted Decision Tree (GBDT) classifier using pretrained contextual embeddings (e.g., BERT embeddings). This baseline will establish the initial magnitude of dialectal bias in a strong non-neural architecture.

Baseline 2: Off-The-Shelf Toxicity Models

We plan to evaluate widely-used pretrained systems such as Toxic-BERT or Google Perspective API to quantify real-world dialect bias in existing moderation systems.

Strategy 1: Vector-Scaling (VS) Loss

We plan to implement the Vector-Scaling loss function within our GBDT architecture. Unlike standard cross-entropy, the VS-loss applies multiplicative and additive factors to logit adjustments specifically for the unprivileged group (AAE).

Strategy 2: Adversarial Gradient Tree Boosting

We will be implementing an Adversarial XGBoost model in which a primary classifier predicts toxicity while a secondary "adversary" model attempts to infer the dialect group from the primary model's leaf outputs. The primary classifier will then be trained to maximize the adversary's prediction error, effectively demoting linguistic confounds.

Experiments & Analysis

Adversarial Fairness Trade-off Sensitivity Analysis: We will conduct a study by varying the adversarial loss weight λ across a range of values. We will analyze how low λ prioritizes toxicity accuracy while allowing bias to persist, versus high λ which aims for near-complete debiasing at the potential cost of overall accuracy. Our goal is to identify the optimal balance between fairness and accuracy.

Hyperparameter Sensitivity Analysis: We will investigate how different regularization strategies affect dialect bias, including L1/L2 regularization and XGBoost-DART (Dropout Additive Regression Trees).

Cross-Dataset Robustness: To test generalization, we will train our fairness-aware model on the noisy TwitterAAE labels and evaluate them on expert-annotated datasets such as SBIC and HateXplain. This experiment will assess whether fairness improvements transfer across datasets with different annotation schemes and linguistic distributions.

Plan for Pre-Final Report and Check-In

- Implementation of baseline strategies on Twitter AAE corpus.
- Successful integration of the Vector-Scaling loss into the training pipeline and initial hyperparameter sweep to evaluate how False Positive Rate moves between dialect groups.
- Generated Baseline Bias Metrics (Accuracy, F1, and Equality of Opportunity Difference) for the standard XGBoost model across both evaluation datasets.

References

Ghimire, S. 2025. How AI Fails: An Interactive Pedagogical Tool for Demonstrating Dialectal Bias in Automated Toxicity Models. *arXiv:2511.06676v1*.